

Practical 10

Write shell script programs for the following on two numbers :

1. addition
2. Subtraction
3. Multiplication
4. Division

```
student@student-virtual-machine:~$ cat > maths5.sh
#maths
echo "enter first number : "
read a
echo "enter second number : "
read b
echo "addition $((a+b))"
echo "subtraction $((a-b))"
echo "multiplication $((a*b))"
echo "division $((a/b))"
^C
student@student-virtual-machine:~$ chmod u+x maths5.sh
student@student-virtual-machine:~$ ./maths5.sh
enter first number :
45
enter second number :
5
addition 50
subtraction 40
multiplication 225
division 9
```

Command to find greatest of two numbers using if else statement

```
student@student-virtual-machine:~$ cat > maths8.sh
echo "enter first value : "
read a
echo "enter second value : "
read b
if [ $a -gt $b ];
then
    echo "$a is greater"
else
    echo "$b is greater"
fi
^C
student@student-virtual-machine:~$ chmod u+x maths8.sh
student@student-virtual-machine:~$ ./maths8.sh
enter first value :
45
enter second value :
200
200 is greater
```

Find greatest between 3 numbers using elif

```
student@student-virtual-machine:~$ cat > maths12.sh
echo "first number : "
read a
echo "second number : "
read b
echo "third number : "
read c
if [ $a -lt $b ]; then
    t=$a; a=$b; b=$t
fi
if [ $a -lt $c ]; then
    t=$a; a=$c; c=$t
fi
if [ $b -lt $c ]; then
    t=$b; b=$c; c=$t
fi
echo "$a > $b > $c"
^C
student@student-virtual-machine:~$ chmod u+x maths12.sh
student@student-virtual-machine:~$ ./maths12.sh
first number :
45
second number :
66
third number :
2
66 > 45 > 2
```

Program to print numbers from 1 to 5 using for loop

```
student@student-virtual-machine:~$ cat > maths14.sh
for ((i=1; i<=5; i++))
do
    echo $i
done
^C
student@student-virtual-machine:~$ chmod u+x maths14.sh
student@student-virtual-machine:~$ ./maths14.sh
1
2
3
4
5
```

Print all the numbers from 10 to 15

```
student@student-virtual-machine:~$ cat > maths15.sh
for ((i=10; i<=15; i++))
do
    echo $i
done
^C
student@student-virtual-machine:~$ chmod u+x maths15.sh
student@student-virtual-machine:~$ ./maths15.sh
10
11
12
13
14
15
```

Find how many users logged in

```
student@student-virtual-machine:~$ cat > see.sh
echo "number of users logged in : " who | wc -l
^C
student@student-virtual-machine:~$ chmod u+x see.sh
student@student-virtual-machine:~$ ./see.sh
1
```

Count how many files and directories present in the current directory

```
student@student-virtual-machine:~$ cat > list_items.sh
echo "Number of files and directories is: " ls -1 | wc -l
^C
student@student-virtual-machine:~$ chmod u+x list_items.sh
student@student-virtual-machine:~$ ./list_items.sh
5
```